Why Banks Need PeX

Payment exceptions are a resolution problem — not simply a transaction-processing problem.

1. BANK OPERATIONS TODAY · *Status Quo*

- **Manual Investigation:** 70%+ of payment breaks remain manual or spreadsheet/macro-driven across operations desks.
- **SME Dependency:** Critical repair knowledge is trapped in individual operators' heads and unshared desktop files.
- **Fragmented Queries:** Operators execute 6–10 disconnected lookups across Core Ledgers, DDA, Sanctions, and Clearing engines for a single break.
- **Coordination Overhead:** Resolution relies on asynchronous email chains, manual ticketing queues, and unstructured chat.
- **Operational Handoffs:** Tier-1/Tier-2 routing queues inflate turnaround time, increasing SLA breach and liquidity penalties.

2. WHY THE PROBLEM IS HARD · *Multi-Dimensional*

- **Payment Rails:** Disparate syntaxes & protocols across SWIFT MT/MX, Fedwire, CHAPS, TARGET2, ACH, and Instant Rails.
- **Exception Types:** 41 distinct break profiles spanning syntax, sanction false-positives, missing LEI/BIC, duplicate pacs.008, and cutoffs.
- **Disjointed Estates:** Payment Hubs, Core DDA, Sanctions screening, and Clearing portals operate as independent silos with distinct lifecycles.

CORE OPERATIONAL REALITY

"An exception rarely belongs to one system,

one rule, or one operational team."

3. THE NEED: COMMON INTELLIGENCE · *Architectural Fit*

- **The Structural Gap:** Payment engines process transactions linearly; they lack contextual intelligence when execution breaks occur.
- **Bridge the Disconnect:** Interpose an intelligent resolution layer between fragmented exception events and bank execution systems.
- **Context Synthesis:** Corroborate historical resolution patterns, peripheral bank evidence, and clearing rail rules into a unified decision.
- **Enterprise Consistency:** Deliver uniform resolution logic across retail, commercial, and correspondent banking business lines.

OPERATIONAL MATURITY ► PEOPLE-DRIVEN (Manual & SME Silos) ► PROCESS-DRIVEN (Task Automation) ► WORKFLOW-DRIVEN (Ticketing) ► RESOLUTION-INTELLIGENCE (Governed Intelligence)

CORE TAKEAWAY: Banks need more than case management or workflow. They need structured resolution intelligence for payment exceptions.

What Is PeX?

A structured Payment Exception Resolution intelligence layer — built around 41 exception profiles.

1. THE 41 EXCEPTION UNIVERSE · *Canonical Scope*

- **Taxonomy Scope:** Spans 41 BIAN-aligned canonical payment exception profiles across 5 operational domains:
- **Account & Routing (12):** Invalid IBAN/BBAN, closed accounts, missing correspondent BIC, routing table mismatch.
- **Data & Format (9):** ISO 20022 XML syntax errors, MT103 field truncation, charge code (OUR/BEN/SHA) errors.
- **Compliance & Sanctions (8):** OFAC 50% disambiguation, PEP false-positive elimination, FATF Travel Rule gaps.
- **Liquidity & Timing (6):** Clearing cutoff breaches, debit cap limits, value-date mismatch, cover delays.
- **Duplicate & Processing (6):** Duplicate UETR tracking, pacs.004 return processing, camt.056 recall handling.

2. WHAT PeX ADDS · *Logical Capabilities*

- **Exception Intelligence:** Ingests raw alerts, parses multi-rail payloads, and derives the root-cause exception profile.
- **Evidence & Context:** Asynchronously queries peripheral bank systems (DDA, Sanctions, Client Master) via read-only ICD adapters.
- **Hypothesis Elimination:** Evaluates and mathematically scores competing resolution hypotheses.
- **Deterministic Rules:** Applies 100% authoritative banking rules before any proposal is submitted.
- **Human Authorization:** Enforces strict Maker-Checker Segregation of Duties ($0 threshold) and Dual-Auth (≥$250k).
- **Merkle Provenance:** Cryptographically seals every decision node into an immutable SHA-256 chain.

3. WHY THIS IS MORE THAN AI · *Enterprise Distinction*

- **Reasoning Scope:** Generic LLMs generate unconstrained text; PeX produces typed, schema-validated ReasoningResult JSON.
- **Domain Centricity:** Generic AI is prompt-centric; PeX is grounded in 41 Canonical Exception Profiles.
- **Governance & Safety:** Generic AI hallucinates; PeX binds AI advice with 100% authoritative deterministic rules.
- **Data Context:** Generic AI uses static prompts; PeX pulls live corroborating evidence via secure bank ICDs.
- **Auditability:** Generic AI offers opaque logs; PeX provides an unbroken cryptographic SHA-256 audit trail.

CANONICAL RESOLUTION LAYER ► 41 EXCEPTION PROFILES ► BANKING ONTOLOGY ► CANONICAL MODEL ► PERIPHERAL EVIDENCE ► COMMON RESOLUTION INTELLIGENCE

CORE TAKEAWAY: PeX is not 'AI for payments.' It is a structured resolution capability built around a defined exception universe.

How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

1. RESOLUTION JOURNEY · *Detection to Reasoning*

- **1. Ingest & Normalize:** Ingests raw MT/MX/ISO break from Payment Hub; normalizes into canonical PaymentExceptionCase.
- **2. Classification:** Maps break to 1 of 41 canonical profiles; derives root cause, missing fields, and failure codes.
- **3. Evidence Retrieval:** Asynchronously queries external systems (DDA status, client directory, SWIFTRef) via read-only ICDs.
- **4. AI Hypothesis:** Model Gateway evaluates candidate repairs and outputs structured ReasoningResult with confidence scores.

2. CONTROLLED DECISION · *Authority & Boundary*

- **Deterministic Rules:** 100% deterministic rule engine validates proposed repair against clearing schemes. Rules ALWAYS override AI.

CORE GOVERNANCE PRINCIPLE

"AI recommends. Controls validate.

Humans authorize. Systems execute."

- **Sanctions Re-Check:** Automated rescreening against watchlists to verify no new compliance risk is introduced.
- **Human-in-the-Loop:** Operator reviews side-by-side evidence, deterministic checks, and AI rationale on an intuitive canvas.
- **Maker-Checker SoD:** Hard $0 threshold (Maker ≠ Checker). Dual-Authorization required for transfers ≥$250k.

3. EXECUTION & OUTCOME · *Settlement & Closure*

- **Contract Generation:** Generates schema-compliant repair message (e.g., ISO 20022 pacs.008 amendment, camt.029 resolution).
- **Clearing Dispatch:** Securely dispatches packet to clearing rail via bank-controlled egress gateway; logs transport ACK (DISPATCHED).
- **Final Settlement:** Case transitions to SETTLED ONLY upon authoritative receipt of pacs.002 ACCP or internal DDA posting.
- **Merkle Audit Sealing:** Cryptographically seals complete 20-step lifecycle into immutable SHA-256 audit log.

GOVERNED LIFECYCLE ► EVENT ► INTELLIGENCE ► DETERMINISTIC CONTROL ► HUMAN AUTHORIZATION ► CONTROLLED EXECUTION ► FINAL SETTLEMENT ► MERKLE AUDIT

CORE TAKEAWAY: PeX closes the loop — from exception detection through investigation, governed decision, controlled execution, and final settlement.

PeX Fits Around the Bank

Non-intrusive integration. Bank-controlled data. Technology-stack neutrality.

1. BANK ECOSYSTEM
Non-Intrusive Fit

- **Zero Disruption:** Wraps around existing core assets without requiring core banking replacement.
- **Payment Hubs:** Integrates with Finastra, ACI, Volante, and FIS hubs.
- **Core Banking:** Connects to SAP, Temenos, Thought Machine, and Mambu DDA ledgers.
- **Clearing Rails:** Interfaces with SWIFT Alliance Gateway, Fedline, and CHAPS.
- **Compliance:** Reads sanctions feeds from Fircosoft, Actimize, and Bridger.

2. INTEGRATION (ICDs)
Pluggable Contracts

- **REST / JSON APIs:** Synchronous case ingestion, status queries, and operator UI.
- **Enterprise MQ:** IBM MQ, MSMQ, RabbitMQ for transactional queue ingestion.
- **Event Streaming:** Apache Kafka, AWS SQS FIFO, GCP Pub/Sub, Azure Service Bus.
- **File / Batch ICDs:** SFTP, ISO 20022 bulk XML, and MT940/MT103 bulk dumps.
- **Standard Contracts:** Decouples PeX domain logic from underlying transport protocols.

3. DATA & AI BOUNDARY
Zero-PII Isolation

- **Token Vault Enclave:** Raw PII (names, IBANs, tax IDs) is tokenized at the bank ingress boundary.
- **KMS Encryption:** Surrogate tokens stored in dedicated FIPS 140-2 Level 3 KMS-encrypted enclave.
- **Model Gateway:** Advisory LLMs process surrogate tokens exclusively—zero raw PII exposure.
- **Data Sovereignty:** Bank payment data is NEVER used to train foundation models.
- **Retention Control:** Model retention policies strictly adhere to bank-approved tenant governance.

4. TECH NEUTRALITY
Multi-Stack Parity

- **Infrastructure:** Mainframe (z/OS), Distributed On-Prem, Private Cloud, or Hybrid.
- **Payment Formats:** SWIFT MT, ISO 20022 MX (pacs, camt, pain), Fedwire FAIM, ACH NACHA.
- **Cloud Parity:** 100% feature parity across AWS, Google Cloud, Microsoft Azure, or On-Prem.
- **Multi-Model AI:** Supports AWS Bedrock, GCP Vertex AI, Azure OpenAI, or Private Self-Hosted LLMs.

ENTERPRISE INTEGRATION ► EXISTING BANK ESTATE ► STANDARDIZED ICDs & ADAPTERS ► PeX RESOLUTION CORE ► BANK-CONTROLLED EXECUTION

CORE TAKEAWAY: PeX adds a resolution capability around the bank's existing technology estate — without requiring technology standardization.

Why PeX Can Be Trusted and Scaled

Governed. Traceable. Reusable across heterogeneous bank environments.

1. EMBEDDED GOVERNANCE
Intrinsic Controls

- **Intrinsic Controls:** Governance is woven directly into the resolution path, not audited post-facto.
- **Maker-Checker SoD:** Strict $0 threshold ensuring the operator who repairs cannot approve.
- **Dual-Authorization:** Transfers ≥$250k require mandatory dual-signoff by two Level-2 supervisors.
- **Policy Guardrails:** Configurable policy rules block STP execution on high-risk sanctions or embargoed rails.

2. FULL TRACEABILITY
Cryptographic Chain

- **11-Stage Hash Chain:** Every break is tracked from Bank Event ➔ Exception ID ➔ Evidence ➔ AI Reasoning ➔ Rules ➔ Human Auth ➔ Final Settlement.
- **SHA-256 Merkle Seals:** Every state transition produces a cryptographically signed provenance node.
- **WORM Compliance:** Audit logs stored in 7-year unalterable WORM object storage.
- **Regulator Ready:** Instant reconstruction of historical decisions during regulatory audits.

3. COMMON ONTOLOGY
Unified Semantics

- **Heterogeneous Banks:** Normalizes disparate legacy schemas from Bank A, B, and C into unified canonical models.
- **Shared Semantics:** Establishes a common resolution vocabulary across retail, corporate, and wealth desks.
- **Cross-Entity Reuse:** Resolution logic authored for one subsidiary is instantly reusable across others.
- **Domain Grounding:** 100% compliant with ISO 20022 and BIAN standard business architectures.

4. ENTERPRISE REUSE
Multi-Entity Scaling

- **Multi-Tenant Scaling:** Scale across global branches, regional payment hubs, and correspondent networks.
- **Continuous Learning:** Approved resolution patterns enrich the bank's canonical knowledge graph without model weight retraining.
- **Sub-Linear Cost:** Dramatically scales payment exception volumes without linear operational headcount growth.
- **Cloud-Agnostic:** Portable across AWS, GCP, Azure, or On-Premises environments with zero lock-in.

ENTERPRISE PROPOSITION ► COMMON INTELLIGENCE ► BANK-CONTROLLED DATA ► DETERMINISTIC GOVERNANCE ► HUMAN ACCOUNTABILITY ► FULL TRACEABILITY ► SCALABLE RESOLUTION

CORE TAKEAWAY: PeX turns fragmented exceptions into a common, governed resolution capability across the entire banking estate.